Orbit Social Media Application — Handbook (PDF-ready)

Product: Orbit — a photo-first social web application

Primary surfaces: Feed, Reels, Messages, Profile (plus authentication)

Tech stack (current codebase): Next.js (App Router), Tailwind CSS, Supabase (Auth + Postgres when configured), Vercel hosting

Document purpose: Explain what Orbit is, why it exists, how it is used, how it was created through staged delivery, what the key features are, how it reaches audiences, what issues were encountered, and what to improve next time.

How to export this document to PDF (recommended)

Option A — Microsoft Word / Google Docs (fastest, best layout control)

1. Open this file in an editor that renders Markdown (VS Code preview, GitHub, etc.).
2. Copy the rendered content into **Word** or **Google Docs**.
3. Insert/replace screenshots from `docs/assets/` as needed (drag images into the doc).
4. Add a **Cover page** + **Table of contents** (Word: References → Table of Contents).
5. Export: **File** → **Save As / Download** → **PDF**.

Option B — Pandoc (automated, optional)

If you install Pandoc locally, you can convert Markdown to PDF with a PDF engine (requires a LaTeX distribution or an engine like `wkhtmltopdf`). This repo does not require Pandoc for development.

Cover page content (paste at top of your PDF)

Title: Orbit Social Media Application — Engineering & Product Handbook

Subtitle: Feed, Reels, Messages, Profile + Authentication (Supabase + Vercel)

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Repository: `balisikh/orbit-social-media-application`

Deployment platform: Vercel

Backend platform: Supabase

Date: April 2026

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1) Executive summary

Orbit is designed as a modern social product with four core modules:

- **Feed** for browsing posts.
- **Reels** for short vertical video.
- **Messages** for direct conversations.

- **Profile** for identity and a portfolio grid of posts.

The application is engineered to run locally for development, and to deploy to production using **Vercel** + **Supabase** public environment variables for client-safe configuration.

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2) What Orbit is (definition)

Orbit is a **dynamic website** (server-rendered routes + client interactions) that provides social-media workflows:

- **Authentication:** sign up and sign in (Supabase email/password when configured).
- **Profile identity:** display name, username/handle, bio, avatar.
- **Content publishing:** posts (including slideshow-style multi-media posts in local mode paths) and reels.
- **Social graph primitives:** follow-related flows exist in the codebase for Supabase-backed usage (depending on migrations and configuration).

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3) Why Orbit exists (purpose)

Product purpose

- Provide a **cohesive** experience across Feed/Reels/Messages/Profile.
- Prioritize **photo-first** consumption patterns (feed + profile grids).
- Support a realistic path to production hosting (Vercel) with a real backend (Supabase).

Engineering purpose

- Keep UI responsive across **mobile, tablet, and desktop** using Tailwind breakpoints (`sm` , `md` , `lg` , ...).
- Keep authentication and session refresh predictable in production.
- Separate **local preview** development behavior from **production** behavior.

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4) How Orbit is used (user journeys)

Journey A — New user onboarding

1. User visits `/auth/signup` .
2. User creates an account (Supabase).
3. User completes profile basics on `/me/edit` .
4. User creates a first post on `/me/post/new` .
5. User sees content on `/feed` and `/me` .

Journey B — Returning user

1. User visits `/auth/login` .
2. User lands on `/feed` .
3. User checks `/reels` and `/messages` .
4. User visits public profile `/u/[handle]` (when handle exists).

Journey C — Password recovery

1. User visits `/auth/forgot-password` .

2. Supabase sends reset email (requires email provider configuration in Supabase).
3. User completes `/auth/update-password` after the callback flow.

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5) Stages and process (create → design → develop → test → deploy)

Stage 1 — Concept and scope

Outputs: feature list, MVP boundaries, success metrics.

Orbit MVP modules captured in this repo: Feed, Reels, Messages, Profile, Auth.

Stage 2 — UX and information architecture

Outputs: route map and primary navigation.

Primary routes (App Router):

- `/feed`
- `/reels`, `/reels/new`
- `/messages`
- `/me`, `/me/edit`, `/me/post/new`
- `/u/[handle]`
- `/auth/login`, `/auth/signup`, `/auth/forgot-password`, `/auth/update-password`, `/auth/callback`

Stage 3 — System design

Outputs: data model direction, auth strategy, deployment strategy.

Key decisions reflected in the codebase:

- **Supabase** for cloud auth and persistence when `NEXT_PUBLIC_SUPABASE_URL` and `NEXT_PUBLIC_SUPABASE_ANON_KEY` are configured.
- **Vercel** for hosting and continuous deployment from GitHub.

Stage 4 — Development

Outputs: implemented UI + API routes + database migrations (SQL under `supabase/migrations/`).

Engineering notes from this project lifecycle:

- Next.js 16 introduced a rename from root `middleware` convention to `proxy` convention for edge/session refresh patterns.
- Responsive rules were tuned for:
- **Portrait single-image posts** vs **slideshow posts** in the feed.
- **Profile grids** across breakpoints.

Stage 5 — Testing

Minimum production test checklist:

- Auth: signup/login/logout/forgot password (as applicable)
- Profile: `/me` and `/me/edit` persistence
- Feed: post creation and rendering
- Reels: list + upload flow (as configured)

- Messages: inbox behavior (mode-dependent)
- Public profile: `/u/[handle]`

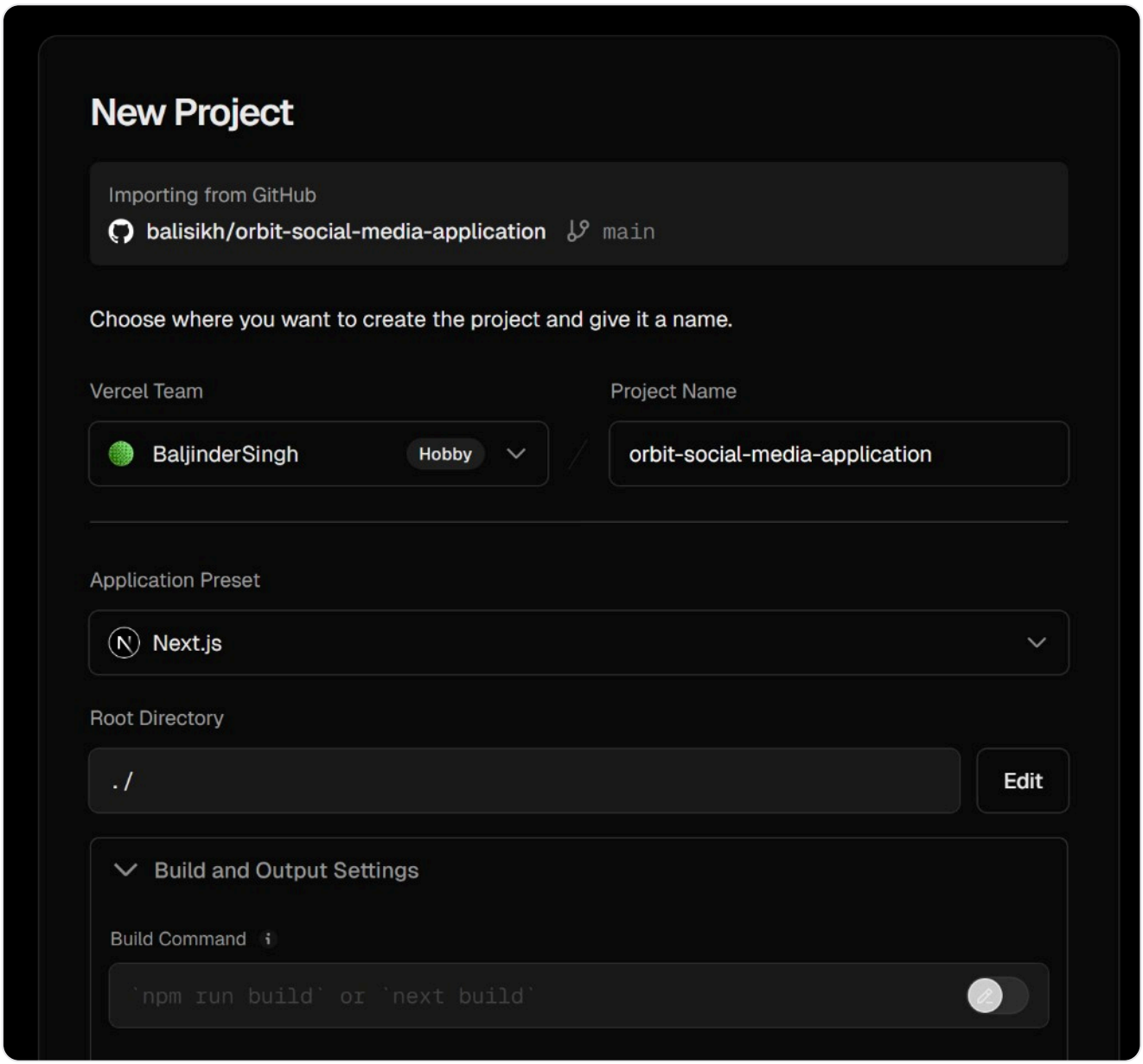
Responsive test widths (examples):

- Mobile: 360×800, 390×844, 412×915
- Tablet: 768×1024, 834×1194, 1024×1366
- Desktop: 1366×768, 1440×900, 1920×1080

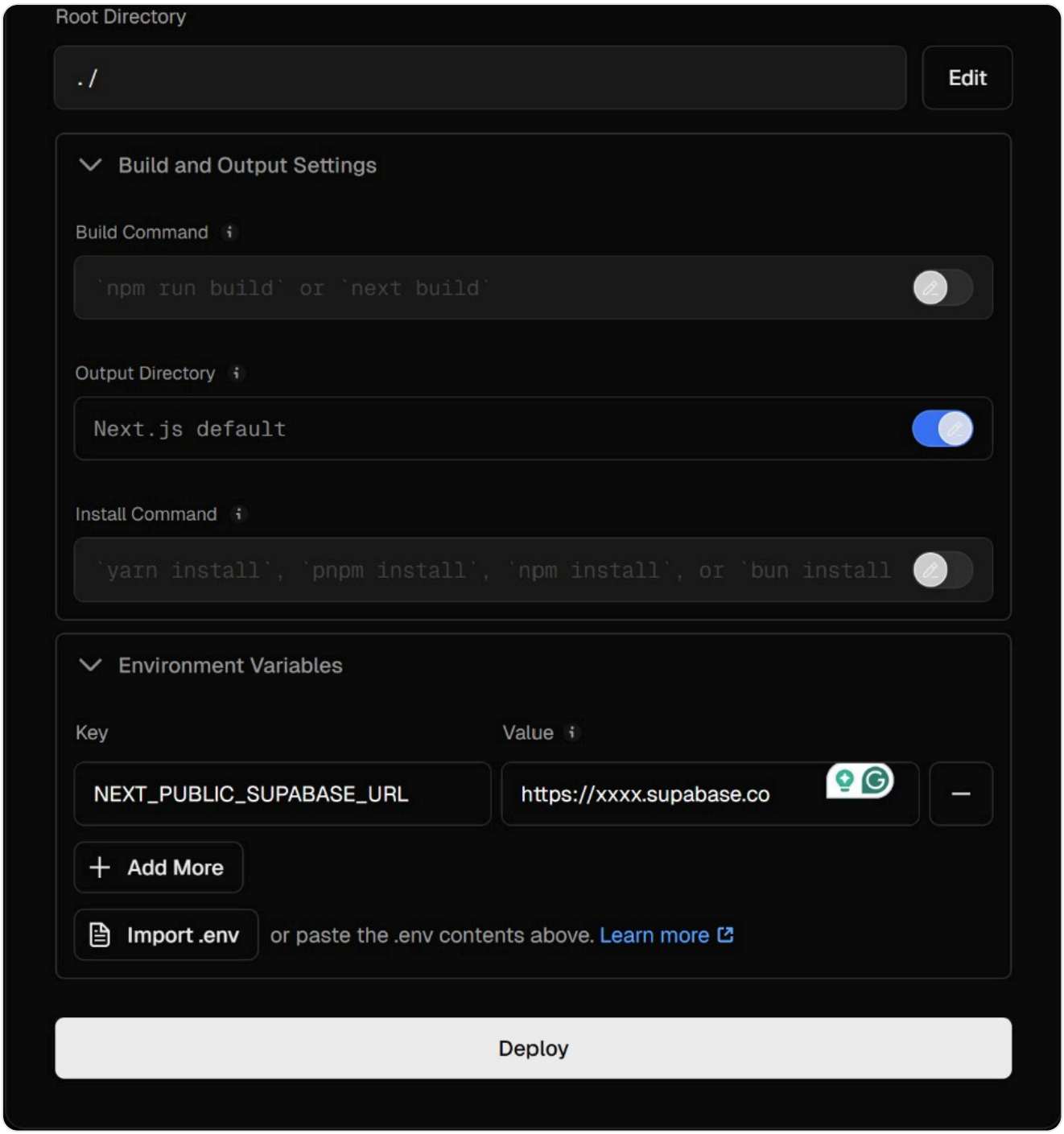
Stage 6 — Deployment (Vercel)

Outputs: live URL, environment configuration, monitoring.

Screenshots included in this handbook (from your deployment flow):



Vercel new project import from GitHub



Vercel new project environment variables and deploy

Important deployment note shown in the screenshot: ensure `NEXT_PUBLIC_SUPABASE_URL` is your real Supabase URL (not a placeholder), and add `NEXT_PUBLIC_SUPABASE_ANON_KEY` as a second variable (not visible in the screenshot, but required by the app).

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6) Key features (what / why / how)

6.1 Feed (`/feed`)

- **What it is:** A scrolling stream of posts.
- **Why it exists:** Primary daily-use surface for social content consumption.

- **How it is used:** Users scroll posts, interact with actions provided by the UI (likes/comments/share vary by mode), and navigate to profile-related flows.

Content-dependent UI behavior (important for documentation accuracy):

- **Portrait photos + short captions:** framed for consistent card layout.
- **Slideshow posts:** framed for swipe/arrow navigation with consistent height on larger breakpoints.

6.2 Reels (`/reels`, `/reels/new`)

- **What it is:** Vertical short video browsing and publishing entry points.
- **Why it exists:** Short video is a distinct consumption pattern from static feed posts.
- **How it is used:** Users browse reels tabs and upload reels.

6.3 Messages (`/messages`)

- **What it is:** Direct messaging UI.
- **Why it exists:** Completes the social loop beyond public posting.
- **How it is used:**
- With Supabase configured: server-backed messaging paths are used.
- Without Supabase (development preview): local browser preview behavior may be used for demos.

6.4 Profile (`/me`, `/me/edit`, `/u/[handle]`)

- **What it is:**
- `/me` is the signed-in owner profile hub.
- `/u/[handle]` is the public profile page for a handle.
- **Why it exists:** Identity + portfolio + shareable profile link.
- **How it is used:** Users set display name, username/handle, bio, avatar; visitors can follow/message depending on configuration.

6.5 Authentication (`/auth/login`, `/auth/signup`, password reset pages)

- **What it is:** Email/password authentication via Supabase when configured.
- **Why it exists:** Secure accounts and server-backed personalization.
- **How it is used:** Users sign up/sign in; sessions are maintained for app routes.

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7) Reaching target audiences (distribution and growth)

Who the product is for (example segments)

- **Creators** who want a simple portfolio + posting flow.
- **Friends/community groups** who want feed + DMs.
- **Early adopters** validating UX before scaling features.

How Orbit can reach audiences

- **Shareable public profiles:** `/u/[handle]` links can be posted externally.
- **Invite loops:** “create account → follow → message” flows.
- **Content marketing:** demo reels/posts as examples in feed.

Benefits to users

- **Fast onboarding:** clear module separation (Feed/Reels/Messages/Profile).
- **Cross-device UX:** responsive layouts tuned for common breakpoints.
- **Production viability:** deployable architecture reduces “prototype-only” risk.

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8) Issues encountered (real examples from this project) + causes + mitigations

Issue 1 — “Password reset is not available…” on production

- **Symptom:** reset page shows unavailable messaging.
- **Cause:** Supabase public environment variables not available in the deployment environment (`NEXT_PUBLIC_SUPABASE_URL` / `NEXT_PUBLIC_SUPABASE_ANON_KEY`).
- **Mitigation:** set both variables in Vercel for Production and redeploy; configure Supabase Auth redirect URLs; configure email delivery in Supabase for actual reset emails.

Issue 2 — “Sign-in is not available on this server.”

- **Symptom:** login cannot proceed in production.
- **Cause:** same as Issue 1: missing Supabase public configuration in production; local preview sign-in is intentionally limited to development environments.
- **Mitigation:** configure Supabase env vars on Vercel + redeploy; verify Supabase Auth settings.

Issue 3 — Profile fields reverting / conflicting with auth metadata defaults

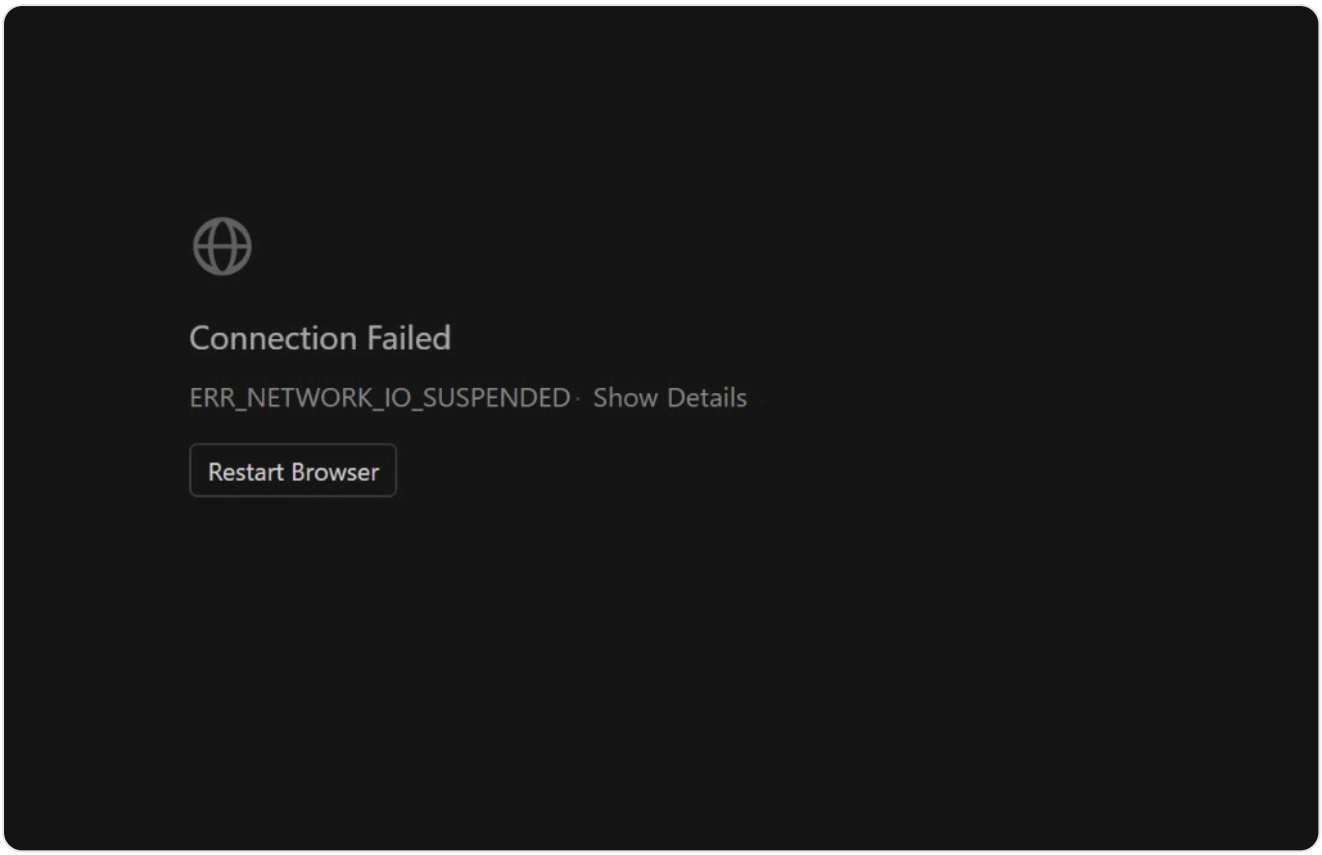
- **Symptom:** profile appears to revert toward email-local defaults after sessions.
- **Cause:** profile sync/merge logic previously favored provider defaults in ways that could overwrite intended profile row values; also confusion between local preview vs cloud mode.
- **Mitigation:** improve merge rules and keep a single “source of truth” strategy per environment; document the difference between preview and production.

Issue 4 — Next.js middleware deprecation warning

- **Symptom:** build warns that middleware file convention is deprecated.
- **Cause:** Next.js 16 renamed the convention to `proxy`.
- **Mitigation:** migrate root `middleware.ts` to `proxy.ts` exporting `proxy`.

Issue 5 — Network connectivity errors during testing

The following screenshot captures a browser-level connectivity failure (`ERR_NETWORK_IO_SUSPENDED`). This is not an Orbit application logic error by itself, but it is a realistic operational issue during local testing or unstable networks.



Connection failed error example

- **Mitigation:** retry on stable network; restart browser; verify VPN/proxy; verify local dev server is running; verify deployed site DNS/SSL on Vercel.

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9) Screenshots you should add for a “complete” PDF (recommended next captures)

The screenshots included above cover **Vercel project creation + env var entry**. For a complete stakeholder PDF, add these additional screenshots (name them consistently under `docs/assets/`):

1. **Supabase dashboard** → **API keys** (blur secrets in PDF if sharing externally)
2. **Supabase** → **Authentication** → **URL configuration** showing redirect URLs
3. **Orbit live site** `/feed`, `/reels`, `/messages`, `/me`, `/me/edit`, `/auth/login`
4. **Vercel deployment success** page showing “Ready” and the live domain

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10) Conclusion — what to do next time (process improvements)

1. **Pre-deploy checklist** pinned in the repo README or team wiki:

- `NEXT_PUBLIC_SUPABASE_URL`
- `NEXT_PUBLIC_SUPABASE_ANON_KEY`
- Supabase redirect URLs include `/auth/callback`
- migrations applied

1. **CI gate:** run `tsc`, `lint`, and `build` on every PR.

2. **Separate "preview vs production" documentation** so testers do not confuse local preview behavior with cloud persistence.
3. **Operational monitoring:** Vercel deployment logs + Supabase auth logs for first-day issues.

Appendix A — Glossary

- **Vercel:** hosting and deployment for the Next.js app.
- **Supabase:** managed Postgres + authentication APIs.
- **RLS:** row-level security policies in Postgres.
- **`NEXT_PUBLIC_*` variables:** bundled into client-side code; must be safe/public values.

Appendix B — File map for screenshots used in this handbook

- `./assets/vercel-new-project-import.png`
- `./assets/vercel-new-project-env-vars.png`
- `./assets/error-connection-failed-network-io-suspended.png`

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